MILITARY SPECIFICATION

FLUXES, WELDING (COMPOSITIONS), SUBMERGED ARC PROCESS WITH TYPE "B" ELECTRODES, CARBON AND LOW-ALLOY STEEL APPLICATION

1. SCOPE

1.1 Scope.- This specification covers granualar welding compositions ("fluxes") for welding carbon and alloy steels and making steel surfacing deposits by the submerged arc process in conjuction with "B" type bare coiled electrodes of Specification MIL-E-18193.

1.2 Classification.- Fluxes shall be furnished in the following types, and particle sizes as specified (see 6.2):

Type	Particle sizes (see 3.4)
MIL-B20	12 by 200, 20 by 200, 20 by D
MIL-B50	8 by 48, 12 by 150
MIL-B80	12 by 65, 20 by 200, 20 by D

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL
UU-S-48 - Sacks, Paper, Shipping.

MILITARY
JAN-S-114 - Sacks, Shipping, Paper, Multiwall (Domestic) (for Other Than Subsistence Items).
MIL-S-16113 - Steel Plate, Hull and Ordnance Structural, Black (Uncoated) and Zinc Coated (Galvanized).
MIL-E-18193 - Electrodes, Welding, Carbon Steel and Carbon-Molybdenum Alloy Steel, Bare Wire, Coiled.

NAVY DEPARTMENT
General Specifications for Inspection of Material.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

FSC 3432

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PUBLICATIONS

BUREAU OF SHIPS
NAVSHIPS 250-692-2 - X-Ray Standards for Production and Repair Weld.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N.Y.)

3. REQUIREMENTS

3.1 Material.- The individual particles of the fluxes shall be homogeneous fused particles containing no free or uncombined silica.

3.2 Uniformity.- Appearance of different samples or lots of the same type of fluxes shall be substantially the same. The quantity of the components specified in 4.3.1 for each flux, when determined by chemical analysis, shall not vary between different parts of the same lot by more than 10 percent from the mean, if the quantity of the component is more than 20 percent, or by more than 20 percent from the mean, if the quantity is less than 20 percent.

3.3 Form.- Fluxes shall be of a free-flowing granular form.

3.4 Sizing.- The particles sizes of the fluxes shall be such that not more than the amounts specified in table I shall be retained on the screen with the larger openings or shall pass through the screen with the smaller openings.

Table I - Particle sizes.

Size Designation	On screen of			Through screen of		
	Maximum	Mesh per inch	Size of square openings	Maximum	Mesh per inch	Size of square openings
	Percent		Inch	Percent		Inch
8 by 48	2.5	8	0.093	3.0	48	0.0116
12 by 65	4.5	12	.055	2.0	65	.0082
12 by 150	8.5	12	.055	2.0	150	.0041
12 by 200	3.5	12	.055	2.0	200	.0029
20 by 200	3.5	20	.0328	4.0	200	.0029
20 by D	3.5	20	.0328	18.0	200	.0029

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3.4.1 Sizings of the various types of fluxes shall be such that they will perform within the welding current ranges specified in table II.

Table II - Standard sizes and working range of welding fluxes.

Current range amperes	Type MIL-B20	Type MIL-B50	Type MIL-B80
Up to 600	12 by 200	8 by 48	12 by 65
600 to 800	12 by 200	8 by 48	12 by 65, 20 by D
800 to 1100	12 by 200, 20 by 200, 20 by D	12 by 150	20 by 200, 20 by D
1100 to 1750	20 by 200, 20 by D	Not recommended	20 by D
Over 1750	20 by D	Not recommended	20 by D

Note.- Except for heavy fillet welding, the first-mentioned size (where two or more are listed) is preferred. The preferred size for heavy fillets and plug welds, even with low currents is 20 by D. This size gives smoother welds, less undercutting, quieter action, wider surface, and reduced penetration.

3.5 Chemical composition.- Chemical composition shall be uniform as specified in 4.3.1.

3.5.1 Distinguishing characteristics.- The different granular fluxes shall be distinguishable from each other by comparison of their colors or by the quantities as determined by analysis of characteristic components specified in table III.

Table III - Characteristic components.

Type	Color	Characteristic component analyzed as	Quantity
			Percent
MIL-B20	Bluish gray	SiO_2	48 minimum
		MnO	2 maximum
MIL-B50	(Finer) dark brown	MnO	38 minimum
	(coarse) black	SiO_2	37 minimum
MIL-B80	Light or sea green	CaO	20 minimum
		Al_2O_3	12 minimum
		Cr_2O_3	1 maximum

3.5.2 Moisture.- Moisture content of all types of fluxes shall not exceed 0.1 percent by weight.

3.6 Performance.-

3.6.1 Appearance.- All fluxes when used as specified in 4.3.4.2 shall result in deposit of weld metal of good contour and penetration, free from surface porosity, cracking, undercutting, or other surface irregularities in excess of those shown on figure 1.

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3.6.2 Soundness. - Weld metal deposited as specified in 4.3.4.3.1 for each type of flux shall not exhibit lack of fusion, cracking, or porosity in excess of that indicated in group I of X-Ray Standards for Production and Repair Welds (NAVSHIPS 250-692-2).

3.6.3 Mechanical properties. - All welding fluxes when tested as specified in 4.3.4.3.2 shall result in deposit of weld metal exhibiting the as-welded mechanical properties specified in table IV.

Table IV - Mechanical properties.

Type	Ultimate strength pounds per square inch	Yield strength pounds per square inch	Elongation 2 inches percent
	Minimum	Minimum	Minimum
MIL-B20	70,000	50,000	22
MIL-B50	85,000	60,000	25
MIL-B80	75,000	53,000	27

4. QUALITY ASSURANCE PROVISIONS

4.1 Acceptance inspection. -

4.1.1 Sampling. -

4.1.1.1 Lot. - For purpose of sampling, and acceptance inspection a lot shall consist of all welding fluxes of one type and size offered for delivery at one time.

4.1.1.2 Sampling for examination of filled containers. - A random sample of filled containers shall be selected by the inspector from each lot in accordance with Standard MIL-STD-105 at inspection level I, acceptable quality level (AQL) equal to 2.5 percent defective, to verify compliance with all the requirements of this specification not involving tests.

4.1.1.3 Sampling for tests. - Two containers shall be selected at random by the inspector from each lot. From each of these containers a 5-pound specimen shall be taken. One of these specimens shall be taken at the time of selection and shall be placed in a separate clean dry metal or glass container sealed and marked for tests, excluding weld test. The other 5-pound specimen may be taken from the second container at the time it is opened for preparation of the required weld. The second 5-pound specimen shall be used to verify test results obtained with the first 5-pound sample in accordance with 4.2.2.

4.2 Acceptance inspection. -

4.2.1 Examination of filled containers. - Each sample filled container selected in accordance with 4.1.1.2, shall be examined by the inspector for defects of the container and closure, for evidence of leakage, and for unsatisfactory markings; each sample filled container shall also be weighed to determine the amount of the contents. Any container in the sample, having one or more defects, or under required fill, shall be rejected. If the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of Standard MIL-STD-105, the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for examination, provided that the contractor has removed or repaired all nonconforming containers.

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4.2.2 Tests, except weld test. - The two 5-pound specimens selected in accordance with 4.1.1.3 shall be subjected to the tests specified in 4.3.1, 4.3.2 and 4.3.3 except that the laboratory may perform complete tests on only one sample from each lot provided sufficient tests are made on the second to verify that both are alike. If one or more tests are failed the lot shall be rejected. Rejected lots may be resubmitted for acceptance tests, provided the contractor has removed or reworked all nonconforming products.

4.2.3 Weld tests. - One of the sample containers selected in accordance with 4.1.1.3 shall be subjected to the performance tests specified in 4.3.4 to verify compliance with 3.6, except where waiver is granted as specified in 4.2.5.

4.2.4 Testing laboratory. - Tests specified in 4.2.2 and 4.2.3 shall be conducted at a laboratory satisfactory to the bureau or agency concerned.

4.2.5 Conditions governing waiver. - X-ray soundness and mechanical property tests may be waived by the bureau or agency concerned if the flux type and size has been previously procured and has passed these tests. A test report of x-ray soundness and mechanical performance shall be furnished.

4.3 Test procedures. -

4.3.1 Uniformity. - A determination of the components as listed in table V for each type of flux, using any standard method yielding results normally accurate within 2 percent, shall not disclose variation greater than specified in 3.2.

Table V - Uniformity.

Type	Components to check for uniformity
MIL-B20	SiO_2, CaO
MIL-B50	SiO_2, MnO
MIL-B80	SiO_2, CaO, Al_2O_3

4.3.2 Sizing. - The sizing shall be determined with standard screens for conformance with 3.4.

4.3.3 Composition. - A determination of the percent by weight of the distinguishing characteristic alloying element specified in 3.5.1, and of the moisture content shall be made by any standard method yielding results normally accurate within 5 percent.

4.3.4 Performance. -

4.3.4.1 Test equipment. - The equipment used in the conduct of performance tests shall be a standard commercial make of automatic welding head, designed for use with and capable of a uniform and controlled feed of granular flux and bare electrode wire; a uniform and adjustable speed of travel; and of sufficient capacity to conduct the required welding current. The head(s) shall be supplied with direct current (d.c.) or alternating current (a.c.) welding power, as required, from a source of adequate capacity to maintain the conditions specified in 4.3.4.2 and 4.3.4.3. Welding current "ground" connection shall be a copper cable lug or lugs of sufficient area of contact that current density shall not exceed 1,000 amperes per square inch, tightly clamped to a smooth, clean, scale-free portion of the plate or plates forming the test weldment. The contact means employed for introducing current into the welding wire electrode shall be such that the area of contact shall be adequate and the location of the area of contact shall not vary during welding. For "constant potential" welding, rate of wire feed shall not change more than plus or minus 4 percent.

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4.3.4.2 Procedure for appearance and operational performance test specimens.- Welds shall be made under the conditions specified in table VI, and in a direction away from the point of the "ground" current connection.

Table VI - Weld test conditions.
(All are butt welds unless noted.) Base metal - grade M of Specification MIL-S-16113.

Type and size	Deposit of granular welding flux		Plate thickness	Vee angle	Unbeveled root face	Electrode type[1]	Electrode diameter	Welding conditions				Type power
	Depth	Width						Amperes	Volts	Speed i.p.m.	Number of passes	
	Inches	Inches	Inches	Degrees	Inch		Inch					
MIL-B20	(Use copper backing bar 1/2 inch by 4 inches with groove 1/2 inch wide by 0.020 inch deep.)											
12 by 200	3/4	2-1/2	1/2	60	3/16	MIL-B1	1/4	850	28	18	1	A.c.
20 by 200	1-1/4	2-3/4	1	35	1/8	MIL-B1	5/16	1550	41	11	1	A.c.
20 by D	1-3/8	4-1/2	2	55	7/8	MIL-B1	3/8	2100	43	6	1	A.c.
MIL-B50	(Use copper backing bar 1/2 inch by 4 inches with no groove.)											
8 by 48	1/2	5/8	14 gage	--	0.078	MIL-B1	3/32	400	21	150	1	D.c. - r.p.
12 by 150	1-1/4	2	5/16	Surface deposit		MIL-B1	1/4	1100	40	32	1	A.c.
MIL-B80	(Use copper backing bar 1/2 inch by 4 inches, for 12 by 66 with no groove, for 20 by 200 and 20 by D with groove 1/2 inch wide and 0.020 inch deep.)											
12 by 66	1/2	5/8	14 gage	0	0.078	MIL-B1	3/32	325	21	100	1	D.c. - r.p.
20 by 200	1	2-1/2	3/4	45	3/16	MIL-B1	1/4	1200	40	12.5	1	A.c.
20 by D	1-1/4	4-1/2	1-1/2	75	5/8	MIL-B1	5/16	1800	43	6.5	1	A.c.

[1]Specification MIL-E-18193.

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4.3.4.2.1 Appearance and operational performance.- Welds made under the conditions specified in 4.3.4.2 shall conform to the appearance requirements specified in 3.6.1 and figure 1.

4.3.4.3 Procedure for making soundness and mechanical properties test specimens.- One pass single vee groove butt welds shall be made in 1/2-inch thick plates in accordance with grade M of Specification MIL-S-16113 prepared with a 60 degree vee and 3/16-inch root face using a copper backing bar 1/2-inch thick by 4-inches wide, with a groove 1/2-inch wide and 0.020-inch deep, and using type MIL-B1 (see Specification MIL-E-18193) wire, 1/4-inch in diameter, operating at from 850 to 900 amperes at approximately 28 volts, with a travel speed of 16 inches per minute. [1] Either d.c. -r.p. (electrode positive) or a.c. welding power shall be used. The granular welding flux shall be deposited approximately 3/4-inch deep above the surface of the plate and 2-1/2-inches wide.

4.3.4.3.1 Soundness.- Soundness of specimens prepared in accordance with 4.3.4.3 shall be determined by radiographs and shall conform to 3.6.2.

4.3.4.3.2 Mechanical properties.- Standard 0.375 inch diameter tensile specimens machined from the fused metal of assemblies fabricated as specified in 4.3.4.3 shall conform with 3.6.3 when tested in a standard tensile testing machine.

4.4 Inspection procedures.- For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

5. PREPARATION FOR DELIVERY

5.1 Packing.-

5.1.1 Level A.- Welding fluxes shall be furnished in 100-pound quantity multi-wall paper bags conforming to Specification UU-S-48. Sacks shall be of the "W" series or "X" series as specified (see 6.2) depending on the destination.

5.1.2 Level B.- Welding fluxes shall be furnished in 100-pound quantity in multi-wall shipping bags conforming to Specification JAN-S-114. One ply shall be of water-resistant materials.

5.1.3 Level C.- Welding fluxes shall be furnished in 100-pound quantity in commercial multi-wall shipping bags having one ply of water-resistant materials. Containers shall be in accordance with the Uniform Freight Classification Rules or other carrier regulations applicable to the mode of transportation.

5.2 Marking.- In addition to any special marking required in the contract or order, or herein, marking for shipment shall be in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use.-

6.1.1 General.- Both the physical properties and the chemical composition of weld metal deposited by the submerged arc process may be influenced by each of the following variables:

(a) Type of granular welding flux.
(b) Electrode composition.
(c) Base metal composition.

As a consequence, 6.1.2 to 6.1.4 should be considered not as specific recommendations, but only as general information. Designations of bare coiled electrodes refer to Specification MIL-E-18193; however, MIL-A1 will give results somewhat comparable to MIL-B3.

[1] These conditions are suitable for 12 by 200 sizes. Welding conditions shall be adjusted to achieve complete penetration: For finer sizes approximately 100 amperes more current shall be used. For coarser sizes up to 5 volts higher voltage and 2 or 3 inches per minute slower speed shall be used.

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6.1.2 Type MIL-B20. - Most generally used for welding steel because of its economy and low sensitivity to surface conditions (such as dirt and scale). This type in the 20 by D size is recommended for heavy fillet welds, and for welds at high currents over 1400 amperes. It may be used for multi-pass welds in structures of medium steel where the joints are not too greatly restrained and welding procedures can be controlled to eliminate cracking. The tensile and impact properties of such weld metal exceed those of the structural quality material being welded. Multi-pass welds made with type MIL-B80 welding flux assure greater freedom from cracking tendencies and improved mechanical properties.

6.1.3 Type MIL-B50. - Recommended for high-speed, high quality welds in gage thicknesses of steel, though it may also be used for joints in heavier thicknesses at currents up to 1100 amperes. This welding flux has the ability to produce sound, leakproof welds on material which, due to ineffective cleaning, may contain mill scale, oil, or other surface residue.

6.1.4 Type MIL-B80. - Designed to produce welds of even better physical properties than obtainable with type MIL-B20 material, and should be used only on material that has been thoroughly cleaned of rust and scale. When molten, this type is quite fluid, and can be used only on flat surfaces or those of relatively little curvature. It should be used for multi-pass welding in preference to other types since it provides suitable limitations of the silicon and manganese contents of the weld metal. It is also used extensively for surface build-up operations.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and particle size of fluxes required (see 1.2).
(c) Level of packing required (see 5.1).
(d) Series of sacks for level A required (see 5.1.1).

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United Stated Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy - Bureau of Ships
(Project 3432-0001Sh)

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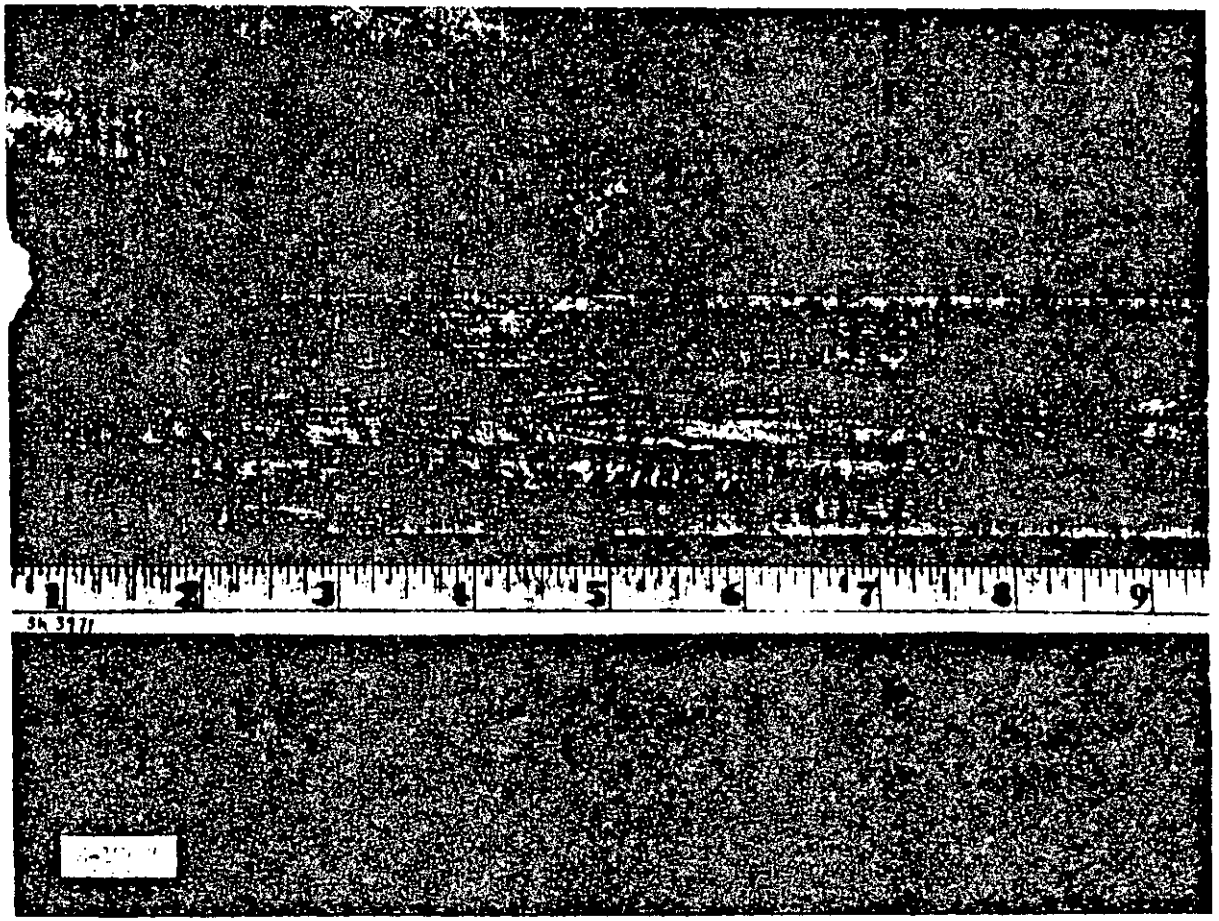


Figure 1 - Appearance standard.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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